

Library-First Activation Steering on Public GPhyT Checkpoints

Public-Only Study

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Abstract

This paper presents a public-only activation-steering study for the General Physics Transformer (GPhyT). The repository now supports checkpoint-backed activation collection, vector fitting, next-step steering sweeps, 20-step autoregressive rollout sweeps, and manuscript asset generation on public Hugging Face checkpoints and public The Well datasets. On the current mean-pressure bootstrap, GPT_S achieves a next-step shift of about $+7.84z$ at steering scale $+2$ on Rayleigh-Benard. Under the new 20-step rollout evaluation, all tested GPT_S/M/L/XL runs remain finite at the evaluated scales, with GPT_XL reaching about $+17.08z$ final pressure shift at $+2$. Transfer to turbulent radiative layer data is strongest for GPT_S, while the larger models remain closer to zero under the same vector and layer selection protocol. A separate regime probe on GPT_S yields validation $AUC = 1.0$ for shear-flow versus Rayleigh-Benard discrimination in the first transformer block.

1 Introduction

GPhyT is a physics foundation model trained across heterogeneous simulation regimes. That makes it a promising target for activation steering: if internal representations already organize physically meaningful quantities, a simple additive intervention might produce controlled shifts in downstream rollouts without retraining the model.

This study is intentionally public-only. It relies only on the public GPhyT checkpoints from Hugging Face and public The Well datasets that can be downloaded locally. The goal is not to reproduce the full private training study, but to establish a reproducible activation-steering pipeline and report what can already be measured from public assets alone.

2 Method

The intervention backend is library-first. The default path uses `pyvene` on native GPhyT transformer blocks, with a thin hook fallback retained for compatibility. All experiments in this paper use transformer block outputs as the intervention sites and additive vector steering as the intervention type.

Feature and regime labels are computed on the final input frame. For scalar features, labels are z-scored within dataset, then reduced to a top-vs-bottom quartile contrast. For regime tasks, labels are taken directly from the dataset identity. The current public experiments use pooled block activations, difference-of-means directions, and logistic-probe normals. The primary cross-model comparisons in this paper focus on the logistic mean-pressure direction fit at `block_out:0`.

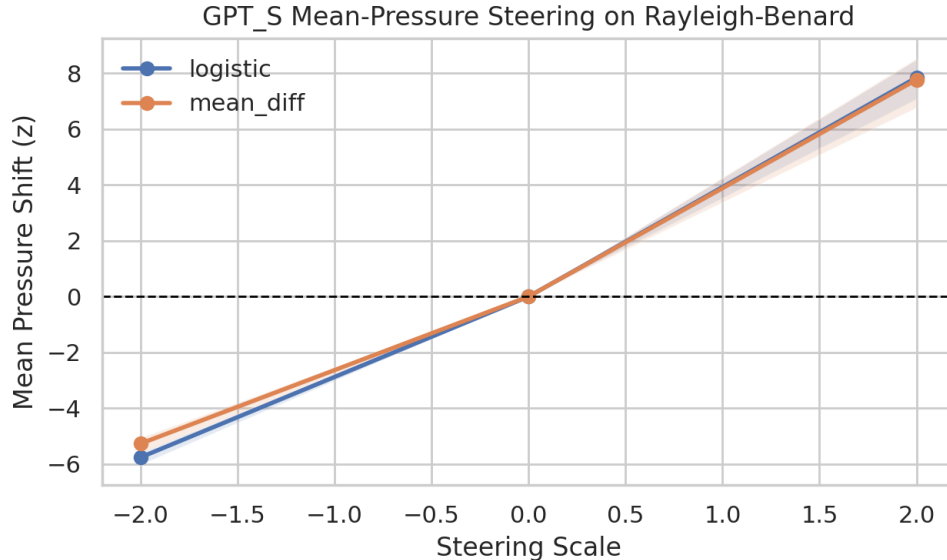


Figure 1: Pressure steering on GPT_S. Both fitted directions produce a monotonic control signal over the tested scale range.

3 Experimental Setup

The current paper uses only public assets:

- public GPhyT checkpoints from Hugging Face,
- public The Well datasets downloaded locally,
- a canonical 5-channel remapping layer inside `PhysicsDataset`, and
- a fixed output resolution matching the public checkpoint geometry.

Because public The Well datasets expose heterogeneous field names and spatial resolutions, the repo standardizes them before they reach the model: `buoyancy` and `tracer` are mapped into the canonical density slot when needed, absent channels are zero-filled, and fields are resized to the checkpoint training resolution.

4 Next-Step Steering

The current bootstrap run uses the public GPT_S checkpoint, the public Rayleigh-Benard dataset, and a single task: steering mean pressure through `block_out:0`. Activations were collected from train and validation splits, directions were fit with both difference-of-means and logistic-probe normals, and a three-point scale sweep was run on the validation split.

The same mean-pressure bootstrap was then repeated for GPT_M, GPT_L, and GPT_XL using the same public dataset, the same first-block intervention site, and the same logistic steering recipe. This is not yet the full study, but it already exposes an important scaling trend in the current public checkpoints: the strongest pressure-control signal in this bootstrap appears in GPT_S, while the

Table 1: Bootstrap steering results for the public GPT_S checkpoint on Rayleigh-Benard validation data.

Method	Scale	Target Shift	Off-Target Drift	Next-Step MSE Delta
mean_diff	-2.000000	-5.265	8.043	0.000769
mean_diff	0.000000	-0.000	0.000	-0.000000
mean_diff	2.000000	7.764	1.313	0.000498
logistic	-2.000000	-5.756	7.895	0.000771
logistic	0.000000	0.000	0.000	0.000000
logistic	2.000000	7.841	1.292	0.000530

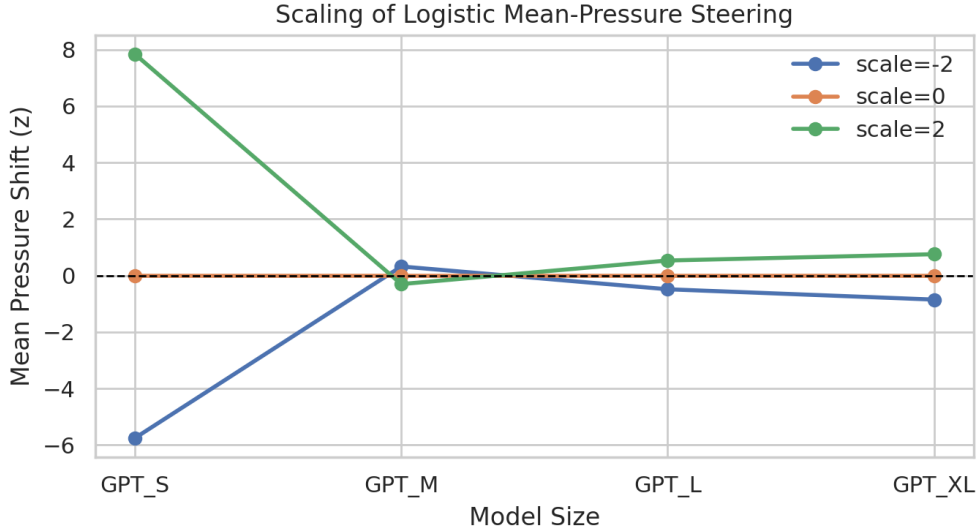


Figure 2: Scaling trend for logistic mean-pressure steering under the current bootstrap protocol. The curves show target feature shift as a function of steering scale for GPT_S/M/L/XL.

larger models trade reduced control magnitude for smaller off-target drift and smaller next-step error deltas.

The bootstrap report already shows strong directional control: negative scales decrease predicted mean pressure, positive scales increase it, and the no-op scale remains numerically stable. The steering effect is large compared with the baseline variation, while next-step MSE deltas remain on the order of 10^{-4} to 10^{-3} for the tested scales.

5 Regime Separation

To move beyond scalar feature steering, a second bootstrap experiment fit a regime-separation direction between public `shear_flow` and `rayleigh_benard` activations using GPT_S at `block_out:0`. Under the current few-batch bootstrap, both the difference-of-means and logistic directions achieved validation AUC = 1.0, indicating that the first transformer block already linearly separates these two regimes in the pooled activation space.

The induced velocity-magnitude shift is small compared with the scalar pressure steering result, but the representation itself is already highly separable. In practice, this suggests that early-

Table 2: Cross-model comparison at steering scale +2 for the logistic mean-pressure direction.

Model	Target Shift @ +2	Off-Target Drift @ +2	Next-Step MSE Delta @ +2
GPT_L	0.532	0.087	0.000019
GPT_M	-0.303	0.158	0.000023
GPT_S	7.841	1.292	0.000530
GPT_XL	0.755	0.218	0.000024

Table 3: Bootstrap regime result for GPT_S. The table reports validation regime-separation AUC and the induced change in mean velocity magnitude at steering scale +2.

Method	Velocity Shift @ +2	Off-Target Drift @ +2	Next-Step MSE Delta @ +2	Validation AUC
mean_diff	-0.0003	0.2036	-0.001795	1.000
logistic	-0.0002	0.2055	-0.001834	1.000

block regime information is present and easy to probe, while turning that information into a large controlled output shift likely requires better layer selection, stronger interventions, or a more regime-specific evaluation objective than the current velocity-magnitude proxy.

6 Autoregressive Rollouts

Next-step control is not enough for a physics model. The steering pipeline now includes 20-step autoregressive rollout sweeps, evaluated on public Rayleigh-Benard validation trajectories with the same logistic mean-pressure vectors used in the next-step study.

The rollout results are qualitatively different from the next-step scaling comparison. GPT_S remains strong, but GPT_XL becomes the strongest rollout controller in the current bootstrap, reaching a final pressure shift of about $17.08z$ at scale +2. All tested models remain finite over the current 20-step evaluation window and zero non-finite failures were observed in the reported sweep.

7 Transfer

To test whether the fitted public vectors transfer beyond their source dataset, the same logistic mean-pressure directions were evaluated on public `turbulent_radiative_layer_2D` validation data.

The transfer behavior is more selective than the in-domain results. GPT_S retains a strong positive control signal ($+2.48z$ at +2), while GPT_M, GPT_L, and GPT_XL remain close to zero under the same intervention recipe. This suggests that the current vector quality is highly dataset-specific for the larger checkpoints, even when in-domain next-step steering looks clean.

8 Discussion

Three observations stand out.

- Raw next-step steerability does not scale monotonically with model size in the current public bootstrap.
- Rollout steerability can differ from next-step steerability: GPT_XL is not the strongest next-step controller here, but it is the strongest 20-step rollout controller.

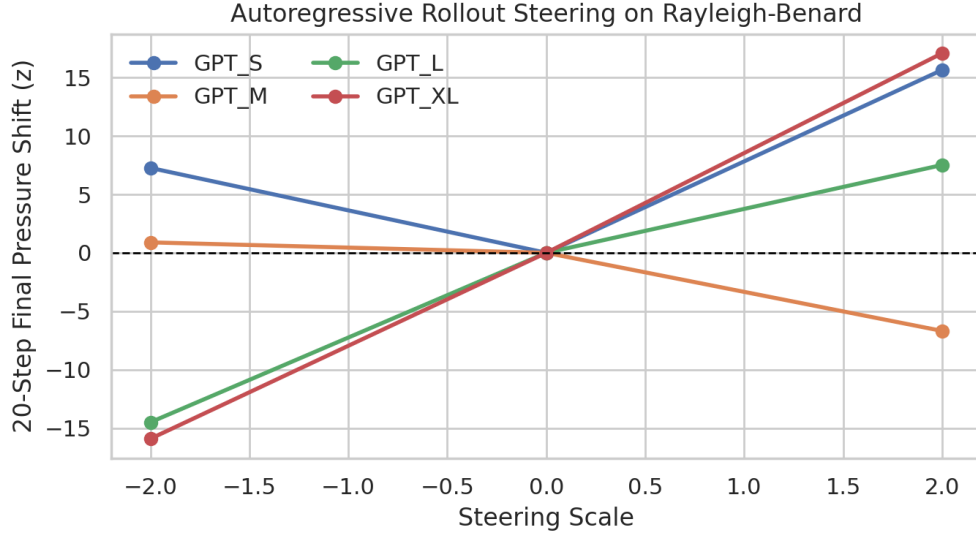


Figure 3: Cross-model 20-step rollout steering on Rayleigh-Benard. The plotted quantity is the final-frame pressure shift in standard deviations relative to the unsteered rollout.

Table 4: 20-step autoregressive rollout summary at steering scale +2.

Model	20-Step Final Shift @ +2	20-Step Final VMSE @ +2	Non-Finite Fraction
GPT_L	7.519	7.336	0.000
GPT_M	-6.683	101.102	0.000
GPT_S	15.652	13.150	0.000
GPT_XL	17.081	9.670	0.000

- Transfer is not guaranteed even when in-domain control is strong. The current public mean-pressure vectors transfer well only for GPT_S.

9 Limitations

This remains a public-only study with a deliberately narrow task family. The current results are based on a single primary scalar feature, one regime probe, and a small number of rollout trajectories. The public `euler_multi_quadrants_periodicBC` dataset was not included in the final paper results because the first attempted local bootstrap download was too large for the writable quota. A fuller study should add derived-physics features, broader layer selection, more transfer datasets, and a more storage-aware bootstrap strategy for the largest public Euler assets.

10 Reproducibility

The repository now contains:

- checkpoint provenance in `artifacts/checkpoint_manifest.json`,

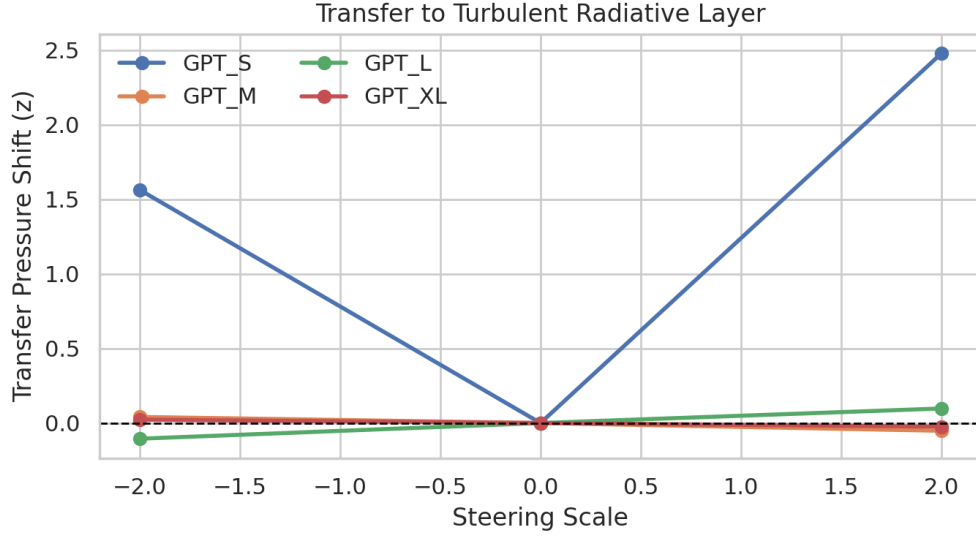


Figure 4: Transfer steering on turbulent radiative layer data. The strongest transferred control signal in the current bootstrap appears in GPT_S.

Table 5: Transfer summary at steering scale +2 on `turbulent_radiative_layer_2D`.

Model	Transfer Shift @ +2	Transfer Drift @ +2	Transfer MSE Delta @ +2
GPT_L	0.097	0.044	-0.000045
GPT_M	-0.052	0.047	-0.000067
GPT_S	2.479	2.073	0.002981
GPT_XL	-0.024	0.220	0.000051

- bootstrap activation collections, vectors, sweeps, and reports under `artifacts/bootstrap/`, and
- manuscript figures and tables generated directly from those reports.